



Frequency Response of the Diffuse-Charge Dynamics in Electrochemical Systems with a Graphene Electrode

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We investigate the frequency response of diffuse-charge dynamics related to a 1:1 symmetric electrolyte containing a graphene electrode by solving the governing Poisson-Nernst-Planck equations subjected to appropriate boundary conditions in the asymptotic limit $\varepsilon = \lambda_D/H \rightarrow 0$, where λ_D is the Debye screening length and H is the half-thickness of the electrolyte. Using the method of matched asymptotic expansion, we first solve the leading order non-linear problem for equilibrium state at a nonzero applied DC voltage in the presence of Stern layer(s). Then, we extend the leading order asymptotic analysis to derive an analytic expression for the impedance of the graphene-based electrochemical cell when a small AC voltage perturbation is added to the applied DC voltage. Finally, we use a suitable scaling of the impedance parameters to expose the impacts of the ion concentration and the DC bias voltage on the frequency response for possible applications involving the graphene-electrolyte interface.

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List of Symbols

D	diffusion constant, [cm ² /sec]
μ	mobility, [cm ² /sec.V]
c_b	bulk ion concentration, [M]
τ	time, [sec]
τ_c	charging time scale, [sec]
τ_H	diffusion time scale, [sec]
τ_D	Debye time scale, [sec]
t	normalized charging time scale
Φ	potential distribution across the cell, [V]
ϕ_g	normalized potential distribution across the GE
ϕ_m	normalized potential distribution across the ME
V_F	potential drop inside graphene, [V]
V_S^g	potential drop across the Stern layer adjacent to the GE, [V]
V_S^m	potential drop across the Stern layer adjacent to the ME, [V]
V_{el}	potential drop across the electrolyte region occupied by the ions, [V]
V	total potential difference applied to the ME with respect to the GE, [V]
v_F	normalized potential drop inside graphene
v_g	potential drop across the EDL adjacent to the GE, [V]
v_m	potential drop across the EDL adjacent to the ME, [V]
v	normalized total potential difference applied to the ME with respect to the GE
k_B	Boltzmann constant, [J/K]
T	Temperature, [K]
e	electron charge, [C]
$C_{\pm}$	positive and negative ion concentrations, [M]
c	normalized ion concentration
$F_{\pm}$	positive and negative ion fluxes, [mol/cm ² .sec]
ϵ_b	dielectric permittivity in the bulk, [μ F/cm]
ϵ_S	dielectric permittivity in the Stern layer, [μ F/cm]
ε	normalized diffuse layer thickness
h	Stern thickness, [nm]
λ_s	effective Stern thickness, [nm]
α	normalized Stern thickness
$\hat{\alpha}$	scaled Stern thickness
σ_g	surface charge density of graphene, [C/cm ²]
ς_g	normalized surface charge density of graphene
ς_d^g	normalized surface charge density in the diffuse layer of the GE

ς_m	normalized surface charge density of metal
ς_d^g	normalized surface charge density in the diffuse layer of the ME
ρ	normalized charge density
n_g	characteristic density of graphene, [nm ⁻²]
u_F	Fermi speed, [m/sec]
n_d	characteristic density in a semi-infinite diffuse layer, [nm ⁻²]
C_S	capacitance of the Stern layer, [μ F/cm ²]
C_D	Debye capacitance, [μ F/cm ²]
C_S	normalized capacitance of the Stern layer
C_d	normalized capacitance of the diffuse layer
C_{DL}	normalized capacitance of the EDL
C	normalized total capacitance
λ_D	Debye length, [nm]
λ_B	Bjerrum length, [nm]
j_c	normalized conduction surface current density
j_d	normalized displacement surface current density
J_{lim}	diffusion limiting current, [A/cm ²]
C_g	normalized quantum capacitance of graphene
γ	coupling ration of graphene
ψ_g	normalized excess potential of the GE relative to the bulk
ψ_m	normalized excess potential of the ME relative to the bulk
s	normalized Laplace-Fourier frequency
ω	frequency, [rad/sec]
$\hat{\omega}$	scaled frequency
$\hat{\omega}_{RC}$	scaled RC frequency
Z	normalized impedance
$\hat{Z}$	scaled impedance
$\mathcal{Z}$	impedance, [Ω]
R_b	normalized bulk resistance
$\mathcal{R}_0$	electrolyte resistance, [Ω]

In numerous exciting applications of graphene, including supercapacitors,^{1,2} biosensors,^{3,4} electrodes,⁵ etc., understanding the graphene-electrolyte interface is essential. The graphene-electrolyte interface can be represented as two capacitors connected in series: the graphene quantum capacitance and the capacitance of the electric double layer (EDL). Based on theoretical calculations, the total interfacial capacitance at low gating potentials is dominated by the quantum capacitance of graphene. This is because graphene has a low density of electronic states at energy levels that are close to the Dirac (or neutrality) point of graphene.^{6,7} The position of the Dirac

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point is determined in a complex way by the interfacial properties of the electrode materials and solvent, which give rise to the potential of zero charge (PZC) in the cell. These properties are discussed in great detail in the 8.

The quantum capacitance of graphene was first experimentally measured by Xia et al.⁹ The measured interfacial capacitance was displayed as a V-shape dependence on externally applied voltage with a non-zero minimum near the Dirac point. Similar voltage-dependence behavior associated with the interfacial capacitance of graphene has been reported recently in different types of electrolytes, including ion-gels¹⁰ and aqueous electrolytes.^{11–13} Little work, however, has been reported regarding the AC capabilities of graphene-electrolyte interface which are critical in the development of related applications such as biosensing.¹⁴

To address this issue, electrochemical impedance spectroscopy (EIS) as well as electrochemical capacitive spectroscopy (ECS) have been extensively used to study the frequency response of the electrode-electrolyte interface^{15–19} because it can provide an approach for characterizing interfacial (bio)physical processes that is fast, sensitive, and label-free. For instance, an EIS biosensor measures the associated impedance by applying a small amplitude AC signal to the electrode over a broad range of frequencies.^{20,21} Then any change in impedance parameters such as phase and magnitude can be used for the detection and biosensing of analytes.²²

In recent years, EIS has gained attention to monitor the biological activities at the surface of the various metal electrodes, such as platinum and Au.²³ However, these electrodes have shown to be problematic in many aspects including insufficient impedance signal, response delay, and impaired sensitivity,^{24–28} with showing unsuitable environmental interface for biosensing activities.²⁹ To alleviate this, graphene electrodes have been suggested as a replacement for metal-electrolyte interfaces.^{30–32} Wang et al.³³ developed an impedimetric biosensor taking advantage of graphene as a biointerface to detect metastatic cancer at the resolution of a single-cell; the response from the graphene biointerface was about 2 times stronger compared to the Au interface. Song et al.³⁴ suggested a graphene-based impedimetric biosensor for drug screening with the response from the graphene biointerface 6.0 times stronger than that of the Au electrode at 54.7 kHz.

Inspired by the aforementioned studies, and to improve upon our earlier work where the frequency response of the uncharged graphene electrode was studied at the Debye relaxation time scale, in this study we will consider that graphene is charged in equilibrium with the diffuse-layer charging dynamics described using the charging relaxation time scale. To achieve this, we perform the leading order asymptotic analysis in the framework of matched asymptotic expansion providing analytic solutions to the statics and diffuse-charge dynamics in the structure of EDLs in response to both finite DC bias and a small AC perturbation of the cell. In addition, we provide more realistic boundary conditions by including Stern layers to model the interplay between the diffuse and compact layers along with the boundary condition pertaining to a single-layer graphene electrode. In the case of a bilayer or a multi-layer graphene electrode, complications arise in the definition of quantum capacitance⁹ due to the substantial changes in the density of states (DOS) of those structures,^{35,36} as well as in the layer-wise doping of those structures by a metallic contact electrode.^{37–39} The latter effect would require a more elaborate boundary condition than the one invoked in our model for single-layer graphene, which goes beyond the scope of the present work.

The paper is organized as follows: In Theory Section, the theory of the concept is explained, beginning with the Poisson-Nernst-Planck (PNP) equations as the governing equations, equipped with a full set of appropriate boundary conditions. With a suitable normalization, the leading order asymptotic analysis of the equilibrium problem through the method of the matched asymptotic expansion is first investigated. Then the time-dependent problem is analyzed in a linear regime with the same asymptotic technique to finally obtain an analytic expression for the impedance of the graphene-based electrochemical cell at leading order. In Results and discussion Section, we discuss the DC and AC behaviors of the present electrochemical cell, taking into account the graphene as an electrode. Conclusions and Future Directions Section offers a final analysis and suggestions for further work.

Theory

The proposed electrochemical cell used in this study contains a symmetrical monovalent binary electrolyte, with two ion species and equal diffusion constants $D_+ = D_- = D$, equal mobilities $\mu_+ = \mu_- = \mu$,

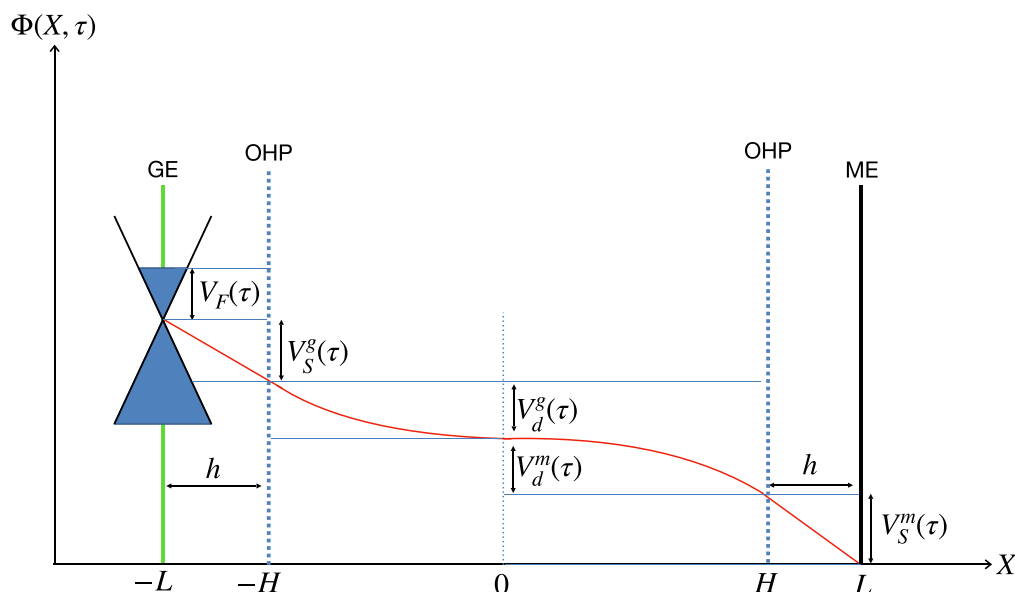


Figure 1. Schematic diagram of the electrochemical cell containing the GE and the ME in an electrolyte bounded by two OHPs near each electrode. As a result of a time-dependent voltage $V(\tau)$, the distribution of electrostatic potential $\Phi(X, \tau)$ is presented with the potential drop $V_F(\tau)$ associated with the doped graphene. The potential decays linearly across the Stern layers extending between each electrode and the corresponding OHP plane, giving rise to the potential drops denoted by $V_S^g(\tau)$ and $V_S^m(\tau)$ respectively. The potential further decays across the adjacent diffuse layers toward a constant value in the bulk region, with the corresponding potential drops denoted by $V_d^g(\tau)$ and $V_d^m(\tau)$.

and identical bulk concentrations c_b in equilibrium. A schematic diagram of the proposed one-dimensional electrochemical cell is provided in Fig. 1, exhibiting its structure along the X axis. In the figure, a graphene electrode (GE) is positioned at $X = -L$ and a metal electrode (ME) at $X = L$, while the electrolyte region is separated from each electrode by a charge-free Stern layer of thickness h . We assume both electrodes to be blocking, neglect specific adsorption of ions and ignore any Faradaic reactions within the Stern layers, so that the electrolyte region extends between the corresponding outer Helmholtz planes (OHPs) at $X = \pm H = \pm L \mp h$, where we impose the zero ion flux boundary conditions. To reveal the role of graphene electrode, we purposefully idealized our model by assuming that the metal electrode has infinite (quantum) capacitance (ideal metal) while neglecting the effects of heterogeneity and atomic steps. This stems from an assumption that the DOS in a metal electrode is so large that charging its surface with excess electrons or holes does not give rise to any appreciable shift of the Fermi energy level in the metal,⁴⁰ resulting in negligible electronic contribution to the system. The charging process on the ME is fully controlled by the potential drop across the electrolyte region occupied by the ions. Moreover, we neglect in our modeling any potential offsets due to differences in the work functions and/or relative electron affinities between different materials, e.g. between graphene and the contact metal electrode used to charge graphene.

As a result of an externally applied time-dependent voltage, $V(\tau)$, an electrostatic potential, $\Phi(X, \tau)$, arises in the cell, with a spatial distribution shown in Fig. 1 along with the potential drops in each region. According to the figure, $V_F(\tau)$ is the potential drop inside graphene linked to the Fermi energy that controls its doping, $V_S^g(\tau) = \Phi(-H, \tau) - \Phi(-L, \tau)$ and $V_S^m(\tau) = \Phi(L, \tau) - \Phi(H, \tau)$ are the potential drops across the Stern layers adjacent to the GE (superscript “g”) and the ME (superscript “m”), respectively. Here, $\Phi(\mp L, \tau)$ denotes the potential at the position of the ME/GE, whereas $\Phi(\mp H, \tau)$ is the potential at the position of the OHP (Stern plane) near those electrodes. Accordingly, the potential drop across the electrolyte region occupied by the ions is given by $V_{el}(\tau) = \Phi(H, \tau) - \Phi(-H, \tau)$, so that the total potential difference applied to the ME with respect to the GE is $V(\tau) = V_F(\tau) + V_S^g(\tau) + V_S^m(\tau) + V_{el}(\tau)$, see Fig. 1.

PNP model.—Using the Einstein’s relation $\mu/D = e/k_B T$, the set of one-dimensional nonlinear PNP equations reads⁴⁰

$$\frac{\partial C_{\pm}}{\partial \tau} = -\frac{\partial F_{\pm}}{\partial X} \quad [1a]$$

$$F_{\pm} = -D \left(\frac{\partial C_{\pm}}{\partial X} \pm \frac{e}{k_B T} \frac{\partial \Phi}{\partial X} C_{\pm} \right) \quad [1b]$$

$$-e_b \frac{\partial^2 \Phi}{\partial X^2} = e(C_+ - C_-) \quad [1c]$$

Here, $C_{\pm}(X, \tau)$ are the positive and negative ion concentrations, $F_{\pm}(X, \tau)$ refers to the corresponding ion fluxes, e_b is the bulk permittivity of the solvent, T is the temperature, k_B is the Boltzmann constant, and $e > 0$ is the proton charge, Eq. 1a, considering Eq. 1b, characterizes the ion transport, and Eq. 1c is the Poisson equation.

The governing PNP equations, Equation 1, need to be solved on the interval $-H < X < H$ subject to the appropriate boundary conditions. Besides the zero-flux conditions at the OHPs for ion concentrations, $F_{\pm}(\pm H, \tau) = 0$, we require two boundary conditions for the potential that include Stern layers. The electric field is constant within the Stern layer adjacent to the GE,

$$\frac{\partial \Phi}{\partial X} \Big|_{X=-L^+} = \frac{\partial \Phi}{\partial X} \Big|_{X=-H^-} \quad [2]$$

while its normal component is continuous across the Stern plane,

$$\epsilon_S \frac{\partial \Phi}{\partial X} \Big|_{X=-H^-} = \epsilon_b \frac{\partial \Phi}{\partial X} \Big|_{X=-H^+} \quad [3]$$

where ϵ_S is the dielectric permittivity in the Stern layer. Recalling that the potential $\Phi(X, \tau)$ is continuous across the cell, the electric field in the Stern layer may also be expressed via the potential difference across that layer according to

$$\frac{\partial \Phi}{\partial X} \Big|_{X=-H^-} = \frac{\Phi(-H, \tau) - \Phi(-L, \tau)}{h} \quad [4]$$

which may be used in Eq. 3 to eliminate the electric field at $X = -H^-$. This gives a non-homogeneous Robin-type boundary condition for the potential at $X = -H^+$,

$$\Phi(-H, \tau) - \lambda_S \frac{\partial \Phi}{\partial X} \Big|_{X=-H^+} = \Phi(-L, \tau) \quad [5]$$

which represents the well-known Stern boundary condition with $\lambda_S = h\epsilon_b/\epsilon_S$ being an effective thickness of the Stern layer.

Following the same procedure, a Stern boundary condition adjacent to the ME, similar to that in Eq. 5, may be derived for the potential at $X = H^-$ as

$$\Phi(H, \tau) + \lambda_S \frac{\partial \Phi}{\partial X} \Big|_{X=H^-} = \Phi(L, \tau) \quad [6]$$

In order to include the GE, we note that its surface charge density (per unit area) is determined by the instantaneous value of the potential $V_F(\tau)$ by using the Dirac cone approximation for graphene’s π electronic bands,

$$\sigma_g(V_F) = en_g [\text{dilog}(1 + e^{\frac{eV_F}{k_B T}}) - \text{dilog}(1 + e^{-\frac{eV_F}{k_B T}})] \quad [7]$$

where n_g is a characteristic density of graphene defined by $n_g = \frac{2}{\pi} \left(\frac{k_B T}{\hbar u_F} \right)^2$, which gives $n_g \approx 9.2 \times 10^{-4} \text{ nm}^{-2}$ at room temperature, with $u_F \approx 10^6 \text{ m/s}$ being the so-called Fermi speed associated with the π bands and dilog the standard dilogarithm function.⁴¹ Applying the Gauss’s law to a “pill box” region encompassing the GE gives

$$-e_S \frac{\partial \Phi}{\partial X} \Big|_{X=-L^+} = \sigma_g(V_F) \quad [8]$$

Using this relation, one can show that the Stern boundary condition adjacent to the GE in Eq. 5 may be replaced by either a non-homogeneous Neumann boundary condition at $X = -H^+$,

$$-e_b \frac{\partial \Phi}{\partial X} \Big|_{X=-H^+} = \sigma_g(V_F) \quad [9]$$

or a non-homogeneous Dirichlet boundary condition for the potential at $X = -H$

$$\Phi(-H, \tau) = \Phi(-L, \tau) - \frac{\sigma_g(V_F)}{C_S}, \quad [10]$$

where $C_S = \epsilon_S/h$ is the capacitance of the Stern layer per unit area. We note that the pill box model introduced in equations 8 and 9 is based on the conventional Gouy-Chapman-Stern (GCS) theory which relates the electrostatic potential at the OHP to the potential in the diffuse layer region. For the modified GCS theory taking into account the finite jellium-like spillover at the electrode/electrolyte interface, a detailed analysis is provided by Kant et al.⁴²

The boundary conditions mentioned above would not produce unique solutions if the charge were not conserved in the system, requiring a constraint to construct a well-posed problem to fully characterize the PNP model in this study. To do so, we take the time derivative of the integrals of both sides in Eq. 1a over the interval $-H \leq X \leq H$, implement the zero-flux conditions at the OHPs, $F_{\pm}(\pm H, \tau) = 0$ and invoke charge neutrality of the electrolyte to obtain

$$\int_{-H}^H C_+(X, \tau) dX = \int_{-H}^H C_-(X, \tau) dX = 2Hc_b. \quad [11]$$

Additionally, integrating the Poisson's equation, Eq. 1c, from $-H^+$ to H^- , gives an expression of the Gauss' law for the electrolyte in the form of an equality of electric fields at the two OHPs,

$$\left. \frac{\partial \Phi}{\partial X} \right|_{X=-H^+} = \left. \frac{\partial \Phi}{\partial X} \right|_{X=H^-} \quad [12]$$

The statements of Eqs. 11 and 12 imply that the charge is conserved in the system and also indicate that the electrodes are coupled, which is found to be a useful relation to demonstrate how the potential distribution connects the two electrodes.

In the following, for simplicity, we drop the superscripts $\pm$ indicating one-sided limits in the boundary conditions without losing the generality of our approach.

Dimensionless formulation.—The problem is reformulated through the reduced variables as follows: the length is scaled to $x = X/H$ and $\ell = L/H$, the time is scaled to $t = \tau/\tau_c$, where $\tau_c = \lambda_D H/D$ is the charging time with λ_D being the Debye length (see below), $c = (C_+ + C_-)/(2c_b)$ is the reduced salt concentration, $\rho = (C_+ - C_-)/(2c_b)$ is the reduced charge density, whereas the reduced potential and the reduced potential drops throughout the cell are defined by $\phi = e\Phi/(k_B T)$ and $v = eV/(k_B T)$, respectively.

The surface charge densities are normalized according to $\varsigma = \sigma/(en_d)$, where $n_d = (4\pi\lambda_B\lambda_D)^{-1}$ is the characteristic density in a semi-infinite diffuse layer, with $\lambda_D = [\epsilon_b k_B T / (2e^2 c_b)]^{1/2}$ being the Debye length and $\lambda_B = e^2 / (4\pi\epsilon_b k_B T)$ the Bjerrum length. As a consequence of such normalization, any capacitance (per unit area) in the system will appear in a dimensionless form that is scaled by the Debye capacitance, $C_D = e^2 n_d / (k_B T) \equiv \epsilon_b / \lambda_D$.

In terms of these variables, the PNP equations, Equation 1, take the dimensionless form for $-1 < x < 1$

$$\frac{\partial c}{\partial t} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial c}{\partial x} + \rho \frac{\partial \phi}{\partial x} \right) \quad [13a]$$

$$\frac{\partial \rho}{\partial t} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial \rho}{\partial x} + c \frac{\partial \phi}{\partial x} \right) \quad [13b]$$

$$\rho = -\varepsilon^2 \frac{\partial^2 \phi}{\partial x^2} \quad [13c]$$

subjected to the following normalized form of the zero-flux and the Stern boundary conditions

$$\left(\frac{\partial c}{\partial x} + \rho \frac{\partial \phi}{\partial x} \right) \Big|_{x=\pm 1} = 0 \quad [14a]$$

$$\left(\frac{\partial \rho}{\partial x} + c \frac{\partial \phi}{\partial x} \right) \Big|_{x=\pm 1} = 0 \quad [14b]$$

$$\phi(-1, t) - \varepsilon \alpha \frac{\partial \phi}{\partial x} \Big|_{x=-1} = \phi(-\ell, t) \quad [14c]$$

$$\phi(1, t) + \varepsilon \alpha \frac{\partial \phi}{\partial x} \Big|_{x=1} = \phi(\ell, t) \quad [14d]$$

where $\alpha = \lambda_S / \lambda_D \equiv C_D / C_S$ is the inverse of the Stern capacitance, normalized by the Debye capacitance. The normalized Neumann and Dirichlet forms of the Stern boundary condition adjacent to the GE in Eq. 14c are obtained from Eqs. 9 and 10 as

$$-\varepsilon \frac{\partial \phi}{\partial x} \Big|_{x=-1} = \varsigma_g(v_F) \quad [15a]$$

$$\phi(-1, t) = \phi(-\ell, t) - \alpha \varsigma_g(v_F) \quad [15b]$$

In the above, the dimensionless parameter $\varepsilon = \lambda_D / H \ll 1$ represents the thickness of a diffuse layer and provides the foundation of asymptotic analysis in this work, whereas the dimensionless parameter α represents the width of the Stern layer relative to the diffuse layer. We note that the limit $\alpha \rightarrow 0$ corresponds to the Gouy-Chapman model, where the Stern layer is ignored while the limit $\alpha \rightarrow \infty$ recovers the Helmholtz model where the diffuse layer vanishes. In this paper, we take into account finite values of α , but always require $\varepsilon \ll 1$.⁴³

The dimensionless total current in the system is $j(t) = j_c(x, t) + j_d(x, t)$, where

$$j_c(x, t) = -\left(\frac{\partial \rho}{\partial x} + c \frac{\partial \phi}{\partial x} \right) \quad [16a]$$

$$j_d(x, t) = -\varepsilon \frac{\partial^2 \phi}{\partial x \partial t} \quad [16b]$$

are dimensionless forms of the conduction current and the displacement current, respectively, which are scaled by the limiting diffusion current, $J_{\text{lim}} = (2eDc_b)/H$.

The system is assumed to be biased by a constant potential $\bar{v}$, giving rise to an equilibrium state, which is further perturbed by a small, time-dependent harmonic potential, $\delta v(t)$, so that

$$v(t) = \bar{v} + \delta v(t) \quad [17]$$

We shall assume that the amplitude of $\delta v(t)$ is small enough, allowing us to work in the first-order perturbation regime that will enable us to linearize the PNP equations and the pertinent boundary conditions and thus calculate the impedance of the biased system. This will be achieved by approximating all the relevant variables with their equilibrium part, indicated by the bar accent, plus a small first-order correction.

For example, by writing $v_F(t) \approx \bar{v}_F + \delta v_F(t)$, the surface charge density of the GE may be approximated by

$$\varsigma_g(v_F) \approx \varsigma_g(\bar{v}_F + \delta v_F) \approx \bar{\varsigma}_g + \delta \varsigma_g \quad [18]$$

where $\bar{\varsigma}_g = \varsigma_g(\bar{v}_F)$ is the dimensionless surface charge density of graphene in equilibrium and $\delta \varsigma_g = -C_g(\bar{v}_F) \delta v_F$ is a small perturbation of that charge density, expressed in terms of the normalized form of the differential quantum capacitance of graphene, defined by

$$C_g(\bar{v}_F) = -\frac{d\bar{\varsigma}_g}{d\bar{v}_F} \quad [19]$$

We note that some details on the derivation of the quantum capacitance are given in the Appendix A of Ref. 40. From Eq. 7, we obtain $C_g(\bar{v}_F) = 2\gamma \ln[2 \cosh(\bar{v}_F/2)]$, where we introduced the ratio $\gamma = n_g/n_d$, which encodes the coupling between the electronic system in graphene and the charge structure of EDLs in the electrolyte.⁴⁴ It is worth mentioning that at the neutrality point, $\bar{v}_F = 0$, the quantum

capacitance of graphene attains its minimum value, given in the reduced form by $C_g^0 \equiv 2\gamma \ln 2$.

Upon substitution of the linearized variables, as in Eq. 17, into the set of Eq. 13a, we first solve the equilibrium problem, followed by a linear perturbation about it, which is then used for calculating the impedance.

Equilibrium solution.—By setting $\partial/\partial t = 0$ in Eq. 13a and integrating the first two equations over x , a system of ordinary differential equations in equilibrium is obtained,

$$\frac{\partial \bar{c}}{\partial x} + \bar{p} \frac{\partial \bar{\phi}}{\partial x} = 0 \quad [20a]$$

$$\frac{\partial \bar{p}}{\partial x} + \bar{c} \frac{\partial \bar{\phi}}{\partial x} = 0 \quad [20b]$$

$$-\varepsilon^2 \frac{\partial^2 \bar{\phi}}{\partial x^2} = \bar{p} \quad [20c]$$

Substituting Eqs. 20c into 20a gives an expression for ion concentration

$$\bar{c}(x) = \bar{c}_0 + \frac{\varepsilon^2}{2} \left(\frac{\partial \bar{\phi}}{\partial x} \right)^2 \quad [21]$$

where $\bar{c}_0 = 1$ is the integration constant, representing the normalized bulk concentration. Plugging Eq. 21 and Eqs. 20c into 20b results in the following master equation for the equilibrium potential

$$\varepsilon^2 \left[\frac{\partial^3 \bar{\phi}}{\partial x^3} - \frac{1}{2} \left(\frac{\partial \bar{\phi}}{\partial x} \right)^3 \right] - \frac{\partial \bar{\phi}}{\partial x} = 0 \quad [22]$$

Owing to the singular perturbations (terms involving ε), Eqs. 21 and 22, do not offer an analytic solution in terms of elementary functions and also pose challenges to solve them numerically. The singular perturbation theory through the method of matched asymptotic expansions provides a convenient solution for each of these challenges. This method evaluates the full PNP model in two distinct domains: the “outer region” or the quasineutral bulk region, and the “inner regions” adjacent to each of the two electrodes, which are associated with diffuse-charge dynamics. As a result, we can obtain an approximate solution that is uniformly valid over the entire cell in the limit $\varepsilon \rightarrow 0$ by appropriately matching the outer and inner regions, giving rise to a model of thin DLs compared to the system size which is valid well below the limiting current density.^{45–47} In the outer region, we seek regular asymptotic expansions of the form $f(x) = f_0(x) + \varepsilon f_1(x) + \varepsilon^2 f_2(x) + \dots$, which is at leading order when $\varepsilon = 0$. Substituting such expansions for $\bar{\phi}(x)$, $\bar{c}(x)$ and $\bar{p}(x)$ into Eq. 22, Eqs. 21 and 20c, along with equating the terms of the same order, we obtain a hierarchy of differential equations for the potential, concentration, and the charge density. At leading order ($\varepsilon = 0$), we find

$$\frac{\partial \bar{\phi}_b}{\partial x} = 0 \quad [23a]$$

$$\bar{c}_b = 1 \quad [23b]$$

$$\bar{p}_b(x) = O(\varepsilon^2) \quad [23c]$$

where the subscript “b” is used to label the outer region as the bulk of electrolyte, to be distinguished from the diffuse layers (inner regions). From Eq. 23a, we can deduce that the potential profile in the bulk region in equilibrium is constant, $\bar{\phi}_b$. Also, Eq. 23c implies the electroneutrality of the outer (bulk) region at leading order.

In the inner region, we magnify the width of the boundary layers by defining the potentials on the side of the GE and the ME as $\bar{\phi}_g(y) = \bar{\phi}(-1 + \varepsilon y)$ and $\bar{\phi}_m(z) = \bar{\phi}(1 - \varepsilon z)$ by means of the stretching transformations for coordinates,

$$y = \frac{1 + x}{\varepsilon} \quad [24a]$$

$$z = \frac{1 - x}{\varepsilon} \quad [24b]$$

respectively. With the coordinates introduced in Eq. 24a, the singular perturbation in the Poisson equation, Eq. 20c, is removed, allowing us to perform the regular asymptotic expansion to obtain the leading order solution for both electrodes separately.⁴⁸

The leading order governing equations for the inner layer on the side of the GE are obtained using Eq. 24a as

$$-\frac{\partial^3 \bar{\psi}_g}{\partial y^3} + \frac{1}{2} \left(\frac{\partial \bar{\psi}_g}{\partial y} \right)^3 + \frac{\partial \bar{\psi}_g}{\partial y} = 0 \quad [25a]$$

$$\bar{c}_g(y) = 1 + \frac{1}{2} \left(\frac{\partial \bar{\psi}_g}{\partial y} \right)^2 \quad [25b]$$

$$\bar{p}_g(y) = -\frac{\partial^2 \bar{\psi}_g}{\partial y^2} \quad [25c]$$

where $\bar{\psi}_g(y) = \bar{\phi}_g(y) - \bar{\phi}_b$ is the excess potential relative to the bulk. The leading order governing equations for the excess potential on the side of the ME, $\bar{\psi}_m(z) = \bar{\phi}_m(z) - \bar{\phi}_b$, and for the corresponding functions $\bar{c}_m(z)$ and $\bar{p}_m(z)$ are the same as in Eq. 25a, except for the subscript g being replaced with m and the variable $y \geq 0$ being replaced with $z \geq 0$.

Transforming Eqs. 14c and 14d to the inner coordinates, the above equations are subjected to the following leading order Stern boundary conditions

$$\bar{\psi}_g(0) - \alpha \frac{\partial \bar{\psi}_g}{\partial y} \bigg|_{y=0} = -\bar{v}_g \quad [26a]$$

$$\bar{\psi}_m(0) - \alpha \frac{\partial \bar{\psi}_m}{\partial z} \bigg|_{z=0} = \bar{v}_m \quad [26b]$$

where $\bar{v}_g = \bar{\phi}_b - \bar{\phi}(-\ell)$ and $\bar{v}_m = \bar{\phi}(\ell) - \bar{\phi}_b$ are the potential drops across the EDLs adjacent to the GE and ME (each consisting of a Stern layer and a diffuse layer), respectively. Referring to Fig. 1, we see that, with the potential in the bulk $\bar{\phi}_b$ being constant, the total externally applied potential difference is decomposed in the equilibrium as $\bar{v} = \bar{v}_F + \bar{v}_g + \bar{v}_m$.

Integrating Eq. 25a, considering the matching condition at the leading order,

$$\lim_{y \rightarrow \infty} \frac{\partial \bar{\phi}_g}{\partial y} \sim \varepsilon \lim_{x \rightarrow -1} \left(\frac{\partial \bar{\phi}_b}{\partial x} \right) = 0 \quad [27]$$

and subsequently letting $\partial^2 \bar{\phi}_g / \partial y^2 = 0$ as $y \rightarrow \infty$, after some algebra we find the leading order electric field in the diffuse layer on the side of the GE

$$-\frac{\partial \bar{\psi}_g}{\partial y} = 2 \operatorname{csch}(y + K_g) \quad [28]$$

and the corresponding leading order excess potential

$$\bar{\psi}_g(y) = 4 \tanh^{-1}(e^{-(y+K_g)}) \operatorname{sign}[\bar{\psi}_g(0)] \quad [29]$$

where K_g is a constant to be determined by the boundary condition at $y = 0$ using Eq. 26a. With the result in Eq. 28 and using the identity

$$\operatorname{csch}(y + K_g) \equiv \sinh\left[\frac{\bar{\psi}_g(y)}{2}\right] \quad [30]$$

which is deduced from the solution in Eq. 29, we observe that Eq. 29 implies the relation

$$-\frac{\partial \bar{\psi}_g}{\partial y} = 2 \sinh\left[\frac{\bar{\psi}_g(y)}{2}\right] \quad [31]$$

Because the governing equations for the inner layer adjacent to the ME are the same as Eq. 25a, an identical leading order electric field is obtained for the ME,

$$-\frac{\partial \bar{\psi}_m}{\partial z} = 2 \sinh\left[\frac{\bar{\psi}_m(z)}{2}\right] \quad [32]$$

Moreover, integrating the Poisson's equation, Eq. 25c, over the range $0 < y < \infty$, along with the matching condition, Eq. 27, and considering Eq. 31, yields the surface charge density in the diffuse layer on the side of the GE as

$$\bar{\zeta}_d^g = \left. \frac{\partial \bar{\psi}_g}{\partial y} \right|_{y=0} = -2 \sinh\left(\frac{\bar{\psi}_g(0)}{2}\right) \quad [33]$$

Following the same procedure and considering Eq. 32, the surface charge density in the diffuse layer on the side of the ME is

$$\bar{\zeta}_d^m = \left. \frac{\partial \bar{\psi}_m}{\partial z} \right|_{z=0} = -2 \sinh\left(\frac{\bar{\psi}_m(0)}{2}\right) \quad [34]$$

With Eqs. 33 and 34, and transforming the coupling boundary condition in Eq. 12 to the inner coordinates,

$$\left. \frac{\partial \bar{\psi}_g}{\partial y} \right|_{y=0} = - \left. \frac{\partial \bar{\psi}_m}{\partial z} \right|_{z=0} \quad [35]$$

we can infer

$$\bar{\psi}_g(0) = -\bar{\psi}_m(0) \quad [36]$$

as well as $\bar{\zeta}_d^g + \bar{\zeta}_d^m = 0$, confirming the charge neutrality in the electrolyte to the leading order in equilibrium. Comparing Eqs. 26a with 26b, from 36 and 35, we find

$$\bar{v}_g = \bar{v}_m \quad [37]$$

which means that the total leading order potential drops across the EDLs adjacent to the GE and ME are equal in equilibrium, so that we may write

$$\bar{v}_F = \bar{v} - 2\bar{v}_g \quad [38]$$

Equation 38 exhibits how the potential drop inside graphene is connected to the total potential drop across both diffuse layers and the applied potential difference across the cell in equilibrium (see Fig. 1).

Upon transforming the normalized Neumann form of the Stern boundary condition in Eq. 15a to the inner coordinate y and taking into account Eq. 33 and Eq. 38, we arrive at the familiar expression for charge neutrality near the GE in equilibrium, i.e.

$\bar{\zeta}_g(\bar{v}_F) + \bar{\zeta}_d^g = 0$, written explicitly as

$$\gamma [\operatorname{dilog}(1 + e^{(\bar{v}-2\bar{v}_g)}) - \operatorname{dilog}(1 + e^{-(\bar{v}-2\bar{v}_g)})] = 2 \sinh\left(\frac{\bar{\psi}_g(0)}{2}\right) \quad [39]$$

Moreover, substituting Eq. 33 into the Stern boundary condition Eq. 26a gives the following relation between the potential drop across the diffuse layer near the GE in equilibrium $\bar{\psi}_g(0)$ and $\bar{v}_g$

$$\bar{\psi}_g(0) + 2\alpha \sinh\left(\frac{\bar{\psi}_g(0)}{2}\right) = -\bar{v}_g \quad [40]$$

We note that the system consisting of Eqs. 39 and 40 can be solved numerically for $\bar{\psi}_g(0)$ and $\bar{v}_g$ to completely characterize the problem in equilibrium for given $\bar{v}$ and α .

Linear dynamics.—While the barred quantities define the equilibrium regime, the linearized governing equations, Eq. 13a, for the perturbations of the ion concentration, δc , charge, $\delta \rho$, and potential, $\delta \phi$, are

$$\frac{\partial \delta c}{\partial t} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial \delta c}{\partial x} + \bar{p} \frac{\partial \delta \phi}{\partial x} + \delta \rho \frac{\partial \bar{\phi}}{\partial x} \right) \quad [41a]$$

$$\frac{\partial \delta \rho}{\partial t} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial \delta \rho}{\partial x} + \bar{c} \frac{\partial \delta \phi}{\partial x} + \delta c \frac{\partial \bar{\phi}}{\partial x} \right) \quad [41b]$$

$$\delta \rho = -\varepsilon^2 \frac{\partial^2 \delta \phi}{\partial x^2} \quad [41c]$$

subjected to the linearized zero-flux and Stern boundary conditions

$$\left(\frac{\partial \delta c}{\partial x} + \bar{p} \frac{\partial \delta \phi}{\partial x} + \delta \rho \frac{\partial \bar{\phi}}{\partial x} \right) \bigg|_{x=\pm 1} = 0 \quad [42a]$$

$$\left(\frac{\partial \delta \rho}{\partial x} + \bar{c} \frac{\partial \delta \phi}{\partial x} + \delta c \frac{\partial \bar{\phi}}{\partial x} \right) \bigg|_{x=\pm 1} = 0 \quad [42b]$$

$$\delta \phi(-1, t) - \varepsilon \alpha \frac{\partial \delta \phi}{\partial x} \bigg|_{x=-1} = \delta \phi(-\ell, t) \quad [42c]$$

$$\delta \phi(1, t) + \varepsilon \alpha \frac{\partial \delta \phi}{\partial x} \bigg|_{x=1} = \delta \phi(\ell, t) \quad [42d]$$

with the perturbations of the conductive and displacement current densities given by

$$\delta j_c(x, t) = - \left(\frac{\partial \delta \rho}{\partial x} + \bar{c} \frac{\partial \delta \phi}{\partial x} + \delta c \frac{\partial \bar{\phi}}{\partial x} \right) \quad [43a]$$

$$\delta j_d(x, t) = -\varepsilon \frac{\partial^2 \delta \phi}{\partial x \partial t} \quad [43b]$$

Denoting the Fourier transform of a perturbed quantity by a tilde, the linearized system in Equation 41 becomes in the frequency domain

$$s \delta \tilde{c} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial \delta \tilde{c}}{\partial x} + \bar{p} \frac{\partial \delta \tilde{\phi}}{\partial x} + \delta \tilde{\rho} \frac{\partial \bar{\phi}}{\partial x} \right) \quad [44a]$$

$$s\delta\tilde{\rho} = \varepsilon \frac{\partial}{\partial x} \left(\frac{\partial \delta\tilde{\rho}}{\partial x} + \bar{c} \frac{\partial \delta\tilde{\phi}}{\partial x} + \delta\tilde{c} \frac{\partial \bar{\phi}}{\partial x} \right) \quad [44b]$$

$$\delta\tilde{\rho} = -\varepsilon^2 \frac{\partial^2 \delta\tilde{\phi}}{\partial x^2} \quad [44c]$$

where $s = i\omega\tau_c$ is a normalized Laplace-Fourier frequency. The above equations are subjected to the zero-flux and Stern boundary conditions in the frequency domain, which follows directly from Equation 42 as

$$\left(\frac{\partial \delta\tilde{c}}{\partial x} + \bar{p} \frac{\partial \delta\tilde{\phi}}{\partial x} + \delta\tilde{\rho} \frac{\partial \bar{\phi}}{\partial x} \right) \bigg|_{x=\pm 1} = 0 \quad [45a]$$

$$\left(\frac{\partial \delta\tilde{\rho}}{\partial x} + \bar{c} \frac{\partial \delta\tilde{\phi}}{\partial x} + \delta\tilde{c} \frac{\partial \bar{\phi}}{\partial x} \right) \bigg|_{x=\pm 1} = 0 \quad [45b]$$

$$\delta\tilde{\phi}(-1, s) - \varepsilon\alpha \frac{\partial \delta\tilde{\phi}}{\partial x} \bigg|_{x=-1} = \delta\tilde{\phi}(-\ell, s) \quad [45c]$$

$$\delta\tilde{\phi}(1, s) + \varepsilon\alpha \frac{\partial \delta\tilde{\phi}}{\partial x} \bigg|_{x=1} = \delta\tilde{\phi}(\ell, s) \quad [45d]$$

Similarly, the current densities in the frequency domain are

$$\delta\tilde{j}_c(x, s) = - \left(\frac{\partial \delta\tilde{\rho}}{\partial x} + \bar{c} \frac{\partial \delta\tilde{\phi}}{\partial x} + \delta\tilde{c} \frac{\partial \bar{\phi}}{\partial x} \right) \quad [46a]$$

$$\delta\tilde{j}_d(x, s) = -\varepsilon s \frac{\partial \delta\tilde{\phi}}{\partial x} \quad [46b]$$

Leading order outer solution in linear regime.—Substituting the Poisson's equations 44c, into 44a and 44b gives

$$s\delta\tilde{c} = \varepsilon \frac{\partial^2 \delta\tilde{c}}{\partial x^2} - \varepsilon^3 \frac{\partial^2}{\partial x^2} \left(\frac{\partial \bar{\phi}}{\partial x} \frac{\partial \delta\tilde{\phi}}{\partial x} \right) \quad [47a]$$

$$-s\varepsilon \frac{\partial^2 \delta\tilde{\phi}}{\partial x^2} = -\varepsilon^2 \frac{\partial^4 \delta\tilde{\phi}}{\partial x^4} + \frac{\partial}{\partial x} \left(\bar{c} \frac{\partial \delta\tilde{\phi}}{\partial x} + \delta\tilde{c} \frac{\partial \bar{\phi}}{\partial x} \right) \quad [47b]$$

Performing the regular asymptotic expansion to Eq. 47a, the linear perturbation of the concentration profile is zero at leading order,

$$\delta\tilde{c}_b(x, s) = 0 \quad [48]$$

Recalling that $\bar{c}_b = 1$ and noticing that $\delta\tilde{c} = 0$ from Eq. 44a and $\delta\tilde{\rho} = 0$ from Eq. 44c, we find from Eq. 47b that $\frac{\partial^2 \delta\tilde{\phi}}{\partial x^2} = 0$ at the leading order. Therefore, the perturbation of the potential in the outer (bulk) region is a linear function

$$\delta\tilde{\phi}_b(x, s) = A(s)x + B(s) \quad [49]$$

for $-1 < x < 1$, where $A(s)$ and $B(s)$ are arbitrary “constants”. Accordingly, the leading order conduction current density in the linear regime is found from Eq. 46a to be position-independent in the bulk region,

$$\delta\tilde{j}_{c,b}(s) = -\frac{\partial \delta\tilde{\phi}_b}{\partial x} = -A(s) \quad [50]$$

whereas the leading order displacement current density in the linear regime vanishes according to Eq. 46b,

$$\delta\tilde{j}_{d,b}(x, s) = 0 \quad [51]$$

Equations 50 and 51 indicate that the linearized total current density in the bulk region is entirely conductive, with the constant $A(s)$ that will be specified below by the time (i.e. frequency) dependent matching conditions.

Conservation of current.—The linearized current densities at leading order, Equation 46, are transformed to the inner region adjacent to the GE using Eq. 24a, giving

$$\delta\tilde{j}_c^g(y, s) = \frac{-1}{\varepsilon} \left(\frac{\partial \delta\tilde{\rho}_g}{\partial y} + \bar{c}_g \frac{\partial \delta\tilde{\phi}_g}{\partial y} + \delta\tilde{c}_g \frac{\partial \bar{\phi}_g}{\partial y} \right) \quad [52a]$$

$$\delta\tilde{j}_d^g(y, s) = -s \frac{\partial \delta\tilde{\phi}_g}{\partial y} \quad [52b]$$

where $\delta\tilde{\phi}_g(y, s) = \delta\tilde{\phi}(-1 + \varepsilon y, s)$, etc. We notice that the total linearized current density is independent of the position⁴⁹ in the inner region, and is accordingly fully determined by its surface value, $\delta\tilde{j}_g^g(0, s) = \delta\tilde{j}_c^g(0, s) + \delta\tilde{j}_d^g(0, s)$. Taking into account the zero-flux boundary condition in Eq. 45b at $x = -1$ and invoking Eq. 24a, we may conclude from Eq. 52a that $\delta\tilde{j}_c^g(0, s) = 0$. Therefore, the total linearized current density in the inner region adjacent to the GE only consists of the displacement current density at the surface of graphene, i.e. at the corresponding Stern plane according to Eq. 2, $\delta\tilde{j}_g^g(0, s) = \delta\tilde{j}_d^g(0, s)$, which is obtained by setting $y = 0$ in Eq. 52b.

Notice that Eqs. 16a and 13b combine into a continuity equation,

$$\frac{\partial \rho}{\partial t} + \varepsilon \frac{\partial j_c}{\partial x} = 0 \quad [53]$$

which when linearized, Fourier transformed and rescaled to the inner coordinate using Eq. 24a gives upon substitution of the Poisson's equation, Eq. 41c,

$$\frac{\partial \delta\tilde{j}_c^g}{\partial y} = s \frac{\partial^2 \delta\tilde{\phi}_g}{\partial y^2} \quad [54]$$

This relation is subjected to the boundary condition $\delta\tilde{j}_c^g(0, s) = 0$ and the matching condition $\delta\tilde{j}_c^g(\infty, s) = \delta\tilde{j}_{c,b}(-1, s) = -A(s)$, according to Eq. 50. Integrating Eq. 54 over y from 0 to ∞ along with the matching condition at the leading order,

$$\lim_{y \rightarrow \infty} \left(\frac{\partial \delta\tilde{\phi}_g}{\partial y} \right) \sim \varepsilon \lim_{x \rightarrow -1} \left(\frac{\partial \delta\tilde{\phi}_b}{\partial x} \right) = 0 \quad [55]$$

results in

$$A(s) = s \frac{\partial \delta\tilde{\phi}_g}{\partial y} \bigg|_{y=0} \quad [56]$$

or equivalently $\delta\tilde{j}_{c,b} = \delta\tilde{j}_d^g(0, s)$.

Equation 56 implies that the boundary layer at leading order serves as a capacitor whose total current density varies in response to the conduction current density from the bulk, $\delta \tilde{j}_{c,b}(x, s) = -A(s)$, and is equivalent to the displacement current density at the surface of the GE. It is worth mentioning that time scales such as diffusion time scale, $\tau_H = H^2/D$, and Debye time scale, $\tau_D = \lambda_D^2/D$, compared to the charging time scale, $\tau_c = \lambda_D H/D$ used here, would not produce the result in Eq. 56 at leading order owing to the imbalance of $O(1)$ quantities in the limit $\varepsilon \rightarrow 0$ and thus may violate the physical content of the matching condition considered equivalently as the continuity of current across the diffuse layer at leading order.⁴⁵

Leading order inner solution in linear regime.—Transforming the Eq. 47a to the inner coordinates adjacent to the GE using Eq. 24a, we obtain the following governing equations in the linear regime

$$s\varepsilon\delta\tilde{c}_g = \frac{\partial^2\delta\tilde{c}_g}{\partial y^2} - \frac{\partial^2}{\partial y^2} \left(\frac{\partial\bar{\psi}_g}{\partial y} \frac{\partial\delta\tilde{\psi}_g}{\partial y} \right) \quad [57a]$$

$$-s\varepsilon \frac{\partial^2\delta\tilde{\psi}_g}{\partial y^2} = -\frac{\partial^4\delta\tilde{\psi}_g}{\partial y^4} + \frac{\partial}{\partial y} \left(\bar{c}_g \frac{\partial\delta\tilde{\psi}_g}{\partial y} + \delta\tilde{c}_g \frac{\partial\bar{\psi}_g}{\partial y} \right) \quad [57b]$$

$$\delta\tilde{\rho}_g = -\frac{\partial^2\delta\tilde{\psi}_g}{\partial y^2} \quad [57c]$$

where we defined a linear perturbation of the excess potential in the diffuse layer of the inner region adjacent to the GE by $\delta\tilde{\psi}_g(y, s) = \delta\tilde{\phi}_g(y, s)$ - $\delta\tilde{\phi}_b(-1, s) = \delta\tilde{\phi}(-1 + \varepsilon y, s) + A(s) - B(s)$ with the use of Eq. 49.

At leading order, the linearized governing equations, Equation 57 are

$$\frac{\partial^2\delta\tilde{c}_g}{\partial y^2} - \frac{\partial^2}{\partial y^2} \left(\frac{\partial\bar{\psi}_g}{\partial y} \frac{\partial\delta\tilde{\psi}_g}{\partial y} \right) = 0 \quad [58a]$$

$$\frac{\partial^4\delta\tilde{\psi}_g}{\partial y^4} - \frac{\partial}{\partial y} \left(\bar{c}_g \frac{\partial\delta\tilde{\psi}_g}{\partial y} + \delta\tilde{c}_g \frac{\partial\bar{\psi}_g}{\partial y} \right) = 0 \quad [58b]$$

$$\delta\tilde{\rho}_g = -\frac{\partial^2\delta\tilde{\psi}_g}{\partial y^2} \quad [58c]$$

Integrating once in Eq. 58a and applying the matching condition for the electric field in equilibrium, Eq. 27, together with

$$\lim_{y \rightarrow \infty} \left(\frac{\partial\delta\tilde{c}_g}{\partial y} \right) \sim \varepsilon \lim_{x \rightarrow -1} \left(\frac{\partial\delta\tilde{c}_b}{\partial x} \right) = 0 \quad [59]$$

yields

$$\frac{\partial\delta\tilde{c}_g}{\partial y} - \frac{\partial}{\partial y} \left(\frac{\partial\bar{\psi}_g}{\partial y} \frac{\partial\delta\tilde{\psi}_g}{\partial y} \right) = 0 \quad [60]$$

Integrating once more and applying the linearized matching conditions for the concentration, $\lim_{y \rightarrow \infty} \delta\tilde{c}_g(y, s) \sim \lim_{x \rightarrow -1} \delta\tilde{c}_b(x, s) = 0$ and the electric field in equilibrium, Eq. 27, produces the linearized concentration profile in the inner (diffuse) layer adjacent to the GE at leading order as

$$\delta\tilde{c}_g(y, s) = \frac{\partial\bar{\psi}_g}{\partial y} \frac{\partial\delta\tilde{\psi}_g}{\partial y} \quad [61]$$

Plugging the above result into Eq. 58b, integrating, and matching the electric field using Eq. 27, yields

$$\frac{\partial^3\delta\tilde{\psi}_g}{\partial y^3} = \left[\bar{c}_g(y) + \left(\frac{\partial\bar{\psi}_g}{\partial y} \right)^2 \right] \frac{\partial\delta\tilde{\psi}_g}{\partial y} \quad [62]$$

Recasting the above equation in terms of the electric field, $\delta\tilde{E}_g(y, s) = -\frac{\partial\delta\tilde{\psi}_g}{\partial y}$, along with the substitution of Eq. 25b and Eq. 28, we obtain a second-order linear differential equation with a non-constant coefficient

$$\frac{\partial^2\delta\tilde{E}_g}{\partial y^2} = [1 + 6\text{csch}^2(y + K_g)]\delta\tilde{E}_g \quad [63]$$

whose solution, which vanishes as $y \rightarrow \infty$, is given by⁵⁰

$$U(y) = 3 \text{csch}(y + K_g) \coth(y + K_g) \quad [64]$$

Thus, the linearized electric field may be written as

$$\delta\tilde{E}_g(y, s) = G(s)U(y) \quad [65]$$

where $G(s)$ is a constant to be determined. Notice that Eq. 65 satisfies the condition $\delta\tilde{E}_g(\infty, s) = 0$, which follows from the matching condition for the electric field at leading order, Eq. 55.

Using Eqs. 28 and 30 we obtain the identity (the details can be found in Ref. 43)

$$2 \text{csch}(y + K_g) \coth(y + K_g) \equiv \frac{\partial^2\bar{\psi}_g}{\partial y^2} \quad [66]$$

which may be used to express the electric field, Eq. 65, in terms of the equilibrium excess potential in the diffuse layer adjacent to the GE with the help of Eq. 31 as

$$\delta\tilde{E}_g(y, s) = \frac{3}{2}G(s) \frac{\partial^2\bar{\psi}_g}{\partial y^2} = \frac{3}{2}G(s) \sinh[\bar{\psi}_g(y)] \quad [67]$$

We may now obtain the excess potential in the linear regime for the inner region adjacent to the GE from Eq. 65 by integrating

$$-\frac{\partial\delta\tilde{\psi}_g}{\partial y} = G(s)U(y) \quad [68]$$

from y to ∞ with the use of Eq. 64. After applying the condition $\delta\tilde{\psi}_g(\infty) = 0$, which follows from the asymptotic matching condition

$$\lim_{y \rightarrow \infty} \delta\tilde{\phi}_g(y, s) = \lim_{x \rightarrow -1} \delta\tilde{\phi}_b(x, s) \quad [69]$$

we find

$$\delta\tilde{\psi}_g(y, s) = 3G(s) \text{csch}(y + K_g) = 3G(s) \sinh \left[\frac{\bar{\psi}_g(y)}{2} \right] \quad [70]$$

Next, we define the linearized excess potential in the diffuse layer of the inner region adjacent to the ME as $\delta\tilde{\psi}_m(z, s) = \delta\tilde{\phi}_m(z, s) - \delta\tilde{\phi}_b(1, s) = \delta\tilde{\phi}(1 - \varepsilon z, s) - A(s) - B(s)$, and notice that the governing equations for the corresponding

linearized electric field, $\delta\tilde{E}_m(z, s) = \frac{\partial\delta\tilde{\psi}_m}{\partial z}$, are the same as the equations for the field on the side of the GE, allowing us to write in analogy to Eq. 67

$$\delta\tilde{E}_m(z, s) = \frac{3}{2}M(s) \sinh[\tilde{\psi}_m(z)] \quad [71]$$

where $M(s)$ is another constant that needs to be determined. By further integrating the above equation from z to ∞ and using the condition $\delta\tilde{E}_m(\infty, s) = 0$ gives

$$\delta\tilde{\psi}_m(z, s) = -3M(s) \operatorname{csch}(z + K_m) = -3M(s) \sinh\left[\frac{\tilde{\psi}_m(z)}{2}\right] \quad [72]$$

By setting $y = 0$ and $z = 0$ in Eqs. 67 and 71, respectively, and invoking the linearized coupling condition, $\delta\tilde{E}_g(0, s) = \delta\tilde{E}_m(0, s)$, along with Eq. 36, we conclude that the integration constant $M(s)$ in Eqs. 71 and 72 must satisfy $M(s) = -G(s)$. If we further apply the Gauss law to the diffuse layer in each of the two inner regions, we deduce that the linear perturbation of the effective surface charge densities in the diffuse layers next to the GE and ME are given by $\delta\tilde{\zeta}_d^g = -\delta\tilde{E}_g(0, s) = -\frac{3}{2}G(s) \sinh[\tilde{\psi}_g(0)]$ and $\delta\tilde{\zeta}_d^m = \delta\tilde{E}_m(0, s) = -\frac{3}{2}G(s) \sinh[\tilde{\psi}_m(0)]$, respectively, thereby confirming the charge neutrality in the electrolyte at leading order in the linear regime, $\delta\tilde{\zeta}_d^g + \delta\tilde{\zeta}_d^m = 0$. At the same time, invoking the linearized form of the Neumann boundary condition for the normalized variables in the inner region near the GE, Eq. 9, which reads $\delta\tilde{E}_g(0, s) = \delta\tilde{\zeta}_g$, we also find that the charge on the GE is fully compensated by the charge accumulated in the nearby diffuse layer at leading order, $\delta\tilde{\zeta}_g + \delta\tilde{\zeta}_d^g = 0$.

By setting $y = 0$ and $z = 0$ in Eqs. 70 and 72, respectively, and using $M(s) = -G(s)$ along with Eq. 36 gives $\delta\tilde{\psi}_m(0, s) = -3M(s) \sinh\left[\frac{\tilde{\psi}_m(0)}{2}\right] = -3G(s) \sinh\left[\frac{\tilde{\psi}_g(0)}{2}\right] = -\delta\tilde{\psi}_g(0, s)$. This relation may be further used in the linearized Stern boundary conditions, Eqs. 45c and 45d, which are written in the inner regions adjacent to the GE and ME electrodes as

$$\delta\tilde{\psi}_g(0, s) + \alpha \delta\tilde{E}_g(0, s) = -\delta\tilde{\zeta}_g \quad [73a]$$

$$\delta\tilde{\psi}_m(0, s) - \alpha \delta\tilde{E}_m(0, s) = \delta\tilde{\zeta}_m \quad [73b]$$

respectively. Here, $\delta\tilde{\zeta}_g = \delta\tilde{\phi}_b(-1, s) - \delta\tilde{\phi}(-\ell, s)$ is the linear perturbation of the potential drop across the EDL (including the Stern layer and the diffuse layer) adjacent to the GE and $\delta\tilde{\zeta}_m = \delta\tilde{\phi}(\ell, s) - \delta\tilde{\phi}_b(1, s)$ is the corresponding potential drop across the EDL adjacent to the ME. By adding the left-hand sides of Eqs. 73a and 73b and recalling the coupling condition $\delta\tilde{E}_g(0, s) = \delta\tilde{E}_m(0, s)$, we conclude that the potential drops across the two EDLs are equal in the linear regime, $\delta\tilde{\zeta}_m = \delta\tilde{\zeta}_g$.

We may finally determine $G(s)$ from Eq. 73a by noticing that $\delta\tilde{E}_g(0, s) = \frac{3}{2}G(s) \sinh[\tilde{\psi}_g(0)] = C_d \delta\tilde{\psi}_g(0, s)$, where $\delta\tilde{\psi}_g(0, s) = 3G(s) \sinh\left[\frac{\tilde{\psi}_g(0)}{2}\right]$ and $C_d = \cosh\left[\frac{\tilde{\psi}_g(0)}{2}\right]$ is the normalized capacitance of the diffuse layer near the GE in equilibrium. Substituting those expressions in Eq. 73a, we obtain $\delta\tilde{\psi}_g(0, s) = -\delta\tilde{\zeta}_g/(1 + \alpha C_d)$. Recalling further that $\delta\tilde{E}_g(0, s) = -\delta\tilde{\zeta}_g^g$, we find from Eq. 73a that the linearized charge density in the diffuse layer near the GE may be expressed in terms of the linearized potential drop across the EDL (consisting of the Stern layer and the diffuse layer) as $\delta\tilde{\zeta}_d^g = C_{DL} \delta\tilde{\psi}_g$, where

$$C_{DL}(\tilde{\psi}_g(0)) = \left[\alpha + \operatorname{sech}\left(\frac{\tilde{\psi}_g(0)}{2}\right) \right]^{-1} \equiv (C_s^{-1} + C_d^{-1})^{-1} \quad [74]$$

is the (normalized) capacitance of the EDL adjacent to the GE, featuring a series connection of the diffuse layer capacitance C_d and the Stern layer capacitance $C_s = 1/\alpha$.

Impedance.—In order to calculate the impedance, we note that the total potential difference between the ME and GE in the linear regime is given by $\delta\tilde{\phi}(\ell, s) - \delta\tilde{\phi}(-\ell, s) = \delta\tilde{\zeta}_m + \delta\tilde{\zeta}_g + \delta\tilde{\zeta}_b$, where the potential drop across the outer (bulk) region in the linear regime follows from Eq. 49 as $\delta\tilde{\zeta}_b = \delta\tilde{\phi}_b(1, s) - \delta\tilde{\phi}_b(-1, s) = 2A(s)$. Thus, the total potential difference, applied to the ME with respect to the GE in the linear regime is $\delta\tilde{V} = \delta\tilde{V}_F + 2\delta\tilde{\zeta}_g + \delta\tilde{\zeta}_b$, where we recall that $\delta\tilde{\zeta}_m = \delta\tilde{\zeta}_g = \delta\tilde{\zeta}_d^g/C_{DL}$. From Eq. 18, we may write $\delta\tilde{V}_F = -\delta\tilde{\zeta}_g/C_g = \delta\tilde{\zeta}_d^g/C_g$, where the normalized quantum capacitance of graphene C_g is defined in Eq. 19.

To be consistent with the measurement models, by taking into account that the potential is applied to the GE with ME being grounded, the total impedance of the system can be defined in the linear regime as

$$Z(\omega) = -\frac{\delta\tilde{V}}{\delta\tilde{j}} = -\frac{\delta\tilde{V}_F + 2\delta\tilde{\zeta}_g}{\delta\tilde{j}_d^g} - \frac{\delta\tilde{\zeta}_b}{\delta\tilde{j}_{c,b}} \quad [75]$$

where in the first term we express $\delta\tilde{j}$ as the displacement current on the GE surface from Eq. 52b, $\delta\tilde{j}_d^g = s\delta\tilde{E}_g(0, s) = -s\delta\tilde{\zeta}_d^g$, while in the second term we express $\delta\tilde{j}$ as the conductive current in the bulk from Eq. 50, $\delta\tilde{j}_{c,b} = -A(s)$, which cancels out the factor $A(s)$ in $\delta\tilde{\zeta}_b = 2A(s)$. As a result, the total dimensionless impedance becomes

$$Z(\omega) = R_b + \frac{1}{sC(\bar{v}, c_b)} \quad [76]$$

where $R_b = 2$ denotes the dimensionless resistance in the bulk region and

$$C(\bar{v}, c_b) = [C_g^{-1}(\bar{v}_F) + 2C_{DL}^{-1}(\tilde{\psi}_g(0))]^{-1} \quad [77]$$

is the dimensionless total capacitance. We note that C_{DL} is defined in Eq. 74 as a function of the potential drop across the diffuse layer near the GE in equilibrium, $\tilde{\psi}_g(0)$, whereas C_g is determined by $\bar{v}_F = \bar{v} - 2\bar{v}_g$. Both $\tilde{\psi}_g(0)$ and $\bar{v}_g$ are obtained by solving the system in Eqs. 39 and 40 for any given total potential difference $\bar{v}$ between the electrodes in equilibrium and for any given ion concentration c_b .

Results and Discussion

We investigate the impedance behavior of the system under the influence of external parameters; ion concentration, total applied potential, and the capacitance of Stern layers. We shall express ion concentration c_b in units of M (mol/liter), pertaining to an aqueous solution containing monovalent ions at room temperature with $\lambda_B \approx 7.2$ Å and $\lambda_D = \lambda_{IM}/\sqrt{c_b}$, where $\lambda_{IM} \approx 3.03$ Å is introduced for convenience as the Debye length at the concentration $c_b = 1$ M. The Debye capacitance is also numerically evaluated as $C_D \approx 229\sqrt{c_b}$ μF/cm². To be consistent with experimental results, the concentrations used in this study are $c_b = 10^{-1}$ M, $c_b = 10^{-2}$ M, $c_b = 10^{-3}$ M, $c_b = 10^{-4}$ M, and $c_b = 10^{-5}$ M along with the total potential difference applied to the ME with respect to the GE, $V = 0$, $V = 0.1$ V, $V = 0.2$ V, $V = 0.3$ V, $V = 0.4$ V, and $V = 0.5$ V. Notice

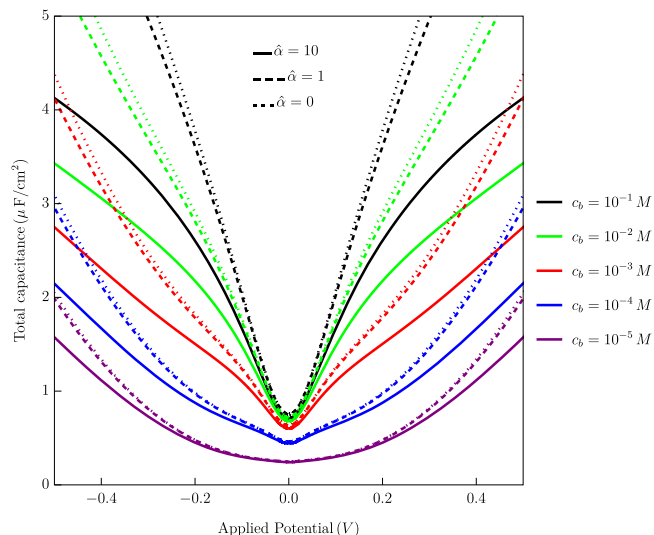


Figure 2. The behavior of the voltage-dependent total capacitance of the system in equilibrium for different values of the ion concentration, c_b , and the scaled Stern layer thickness, $\hat{\alpha}$.

that, while the charge on the GE is nominally zero when $V = 0$, this value does not correspond to the actual PZC in the cell because we neglected in our modeling any offsets in the potential drops across various interfaces between different media. To combine the effects of the Stern layer and the ion concentration, we scale its thickness according to $\alpha = \hat{\alpha}\sqrt{c_b}$ where $\hat{\alpha} = (\epsilon_b/\epsilon_S) \times (h/\lambda_{1M})$. In our numerical calculations, we use $\hat{\alpha} = 10$ and $\hat{\alpha} = 1$ as high- and low-value estimates, along with the case where the Stern layer vanishes ($\hat{\alpha} \rightarrow 0$) by setting $\hat{\alpha} = 10^{-4}$ in our computations.

Equilibrium analysis.—Figure 2 shows the behavior of the total capacitance of the system in equilibrium when subjected to the externally applied potential for different values of ion concentration and the scaled Stern thickness $\hat{\alpha}$. As confirmed by recent experiments,⁵¹ in electrochemical systems containing GE, the total capacitance of the system is dominated by the quantum capacitance of the graphene at high ion concentrations and thus the case with $c_b = 0.1$ M and $\hat{\alpha} = 0$ in Fig. 2 is very close to the voltage-dependent quantum capacitance, exhibiting a minimum value of $0.8 \mu\text{F}/\text{cm}^2$ at $V = 0$. According to the figure, the curves are all concave shaped and tend to group together at the PZC, $V = 0$, where they exhibit a minimum value that decreases with decreasing ion concentration. Also, from the figure, it is evident that at low applied voltages, $|V| \lesssim 0.1$ V, the Stern layer produces no prominent impact on the

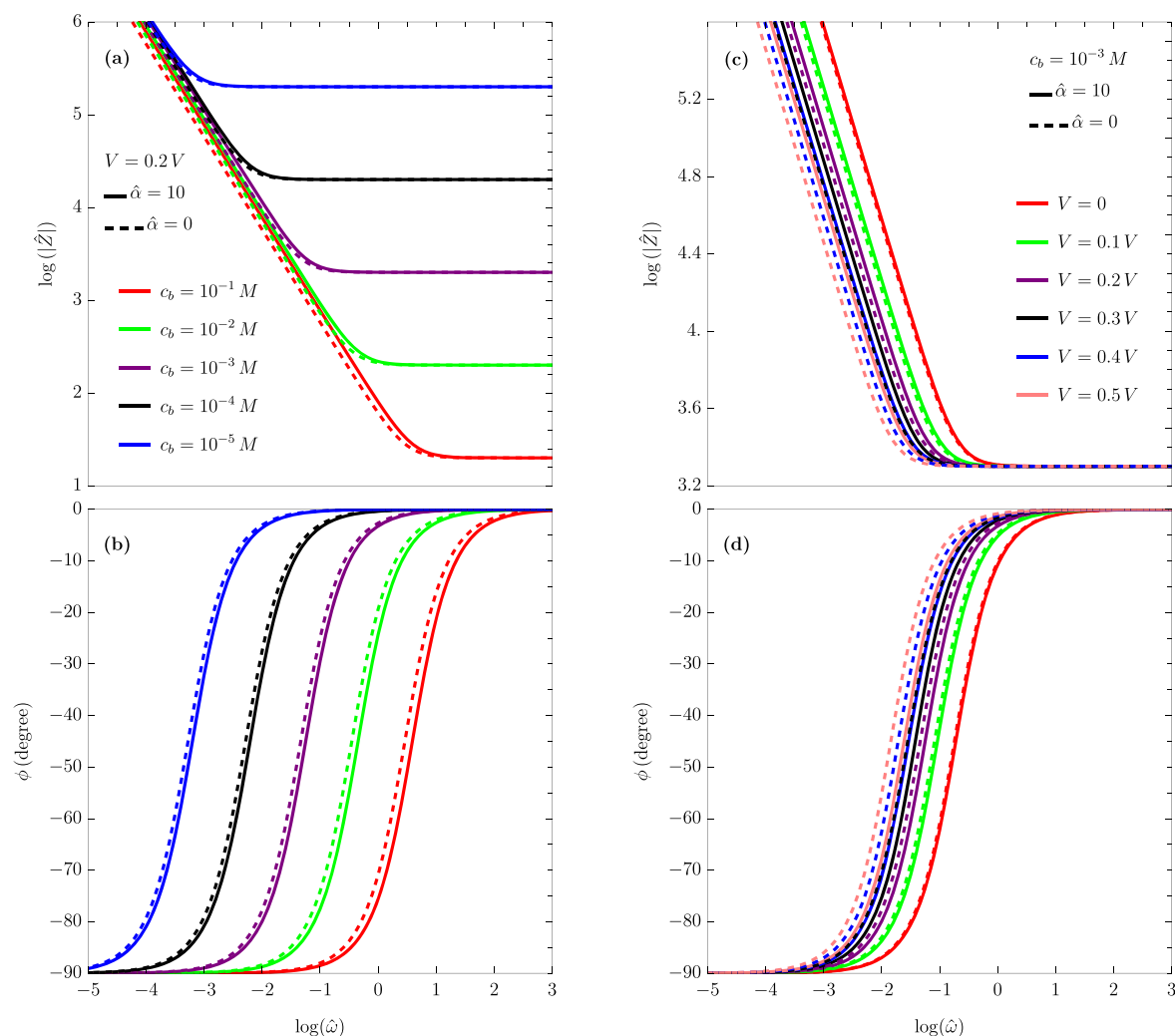


Figure 3. Frequency response of the graphene-based electrochemical cell. Left: the frequency dependence of (a) magnitude and (b) phase angle for different values of the ion concentration, c_b , and the Stern thickness, $\hat{\alpha}$, with the applied potential held at $V = 0.2$ V. Right: the frequency dependence of (c) magnitude and (d) phase angle for different values of the bias voltage and the Stern thickness, $\hat{\alpha}$, with the ion concentration held at $c_b = 10^{-3}$ M.

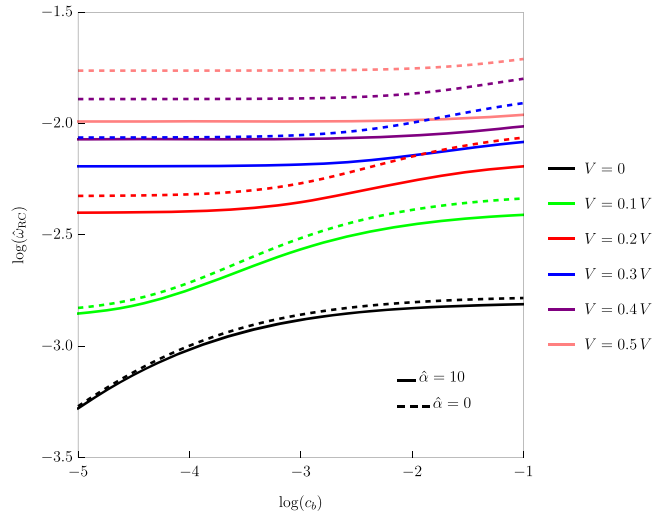


Figure 4. The variation of the characteristic frequency, $\hat{\omega}_{RC}$ vs ion concentration for different values of the bias voltage, V , and the Stern layer thickness, $\hat{\alpha}$.

overall capacitance. However, at high voltages, $|V| \gtrsim 0.1$ V, as the Stern thickness extends to $\hat{\alpha} = 10$, the curves tend to flatten, imposing an upper bound on the overall capacitance. In all cases, the curves with $\hat{\alpha} = 1$ are very close to those without the Stern layer ($\hat{\alpha} = 0$), so we shall only consider the cases $\hat{\alpha} = 10$ and $\hat{\alpha} = 0$ in the following discussion.

Frequency response analysis.—To observe the frequency response of the system, we scale the impedance reported in Eq. 76 according to $Z(\omega) = \mathcal{Z}/\mathcal{R}_0$ where $\mathcal{Z}$ is the impedance in physical units of Ω and $\mathcal{R}_0 = H/(2e\mu A c_b) \equiv \mathcal{R}_{1M}/c_b$ is the resistance (in units of Ω) in which $\mathcal{R}_{1M} = H/(2e\mu A c_{1M})$ refers to the resistance in electrolyte with the ion concentration $c_{1M} = 0.602 \text{ nm}^{-3}$ and the electrode area A . With this definition, we can introduce a new scaling for the impedance as $\hat{Z} = \mathcal{Z}/\mathcal{R}_{1M}$ that is independent of ion concentration, leading to

$$\hat{Z} = \frac{2}{c_b} + \frac{1}{i\hat{\omega}\sqrt{c_b}C(\bar{v}, c_b)} \equiv \hat{Z}_r + i\hat{Z}_i \quad [78]$$

where $\hat{\omega} = \omega\lambda_{1M}H/D$ is introduced as the new scaling for the frequency and $\hat{Z}_r$ and $\hat{Z}_i$ denote the real part and imaginary part of the scaled impedance $\hat{Z}$ respectively. We note that a typical value for the scaling factor in $\hat{\omega}$ can be deduced by choosing $H = 1 \mu\text{m}$ and $D = 10^{-6} \text{ cm}^2/\text{sec}$, which gives $\lambda_{1M}H/D \sim 3 \mu\text{ sec}$. The frequency response is then characterized by the magnitude response and the phase response obtained according to $|\hat{Z}| = \sqrt{\hat{Z}_r^2 + \hat{Z}_i^2}$, and $\hat{\phi} = \arctan(\hat{Z}_i/\hat{Z}_r)$, respectively. In the following we study the frequency response of the system under the influence of external parameters: the ion concentration and the bias voltage, with ($\hat{\alpha} = 10$) and without ($\hat{\alpha} = 0$) Stern layer.

Figure 3 depicts in panels (a) and (b) the frequency response for different values of ion concentration with Stern thicknesses $\hat{\alpha} = 10$ and $\hat{\alpha} = 0$, while the applied potential is held at $V = 0.2$ V. From Fig. 3b, we observe that the phase response approaches -90° as the frequency decreases, corresponding to the slope of -1 in the magnitude plot shown in Fig. 3a. This reflects the fact that the system shows a pure capacitive behavior in the low-frequency regime. However, the phase angle approaches zero as the frequency increases, corresponding to the zero slope in the magnitude plot where curves tend to be horizontal, which is an indication of a pure resistive behavior in the high-frequency regime. Moreover, from

Figs. 3a, 3b, we note that reducing the ion concentration from $c_b = 10^{-1}$ M to $c_b = 10^{-5}$ M, shifts the curves to lower frequencies where the capacitive behavior sets in, making the proposed graphene-based electrochemical cell a potential candidate in (bio) sensing applications where low-frequency responses are highly desired.⁵² Also, in Figs. 3a, 3b we observe negligible effects of the presence of the Stern layer at all ion concentrations.

The ion concentration in the range of mM has been shown to provide an optimal balance between sensitivity and specificity in detecting biological analytes and monitoring cellular processes for ion-selective electrodes.^{53–55} To take advantage of this range of ion concentration, we investigate in the panels (c) and (d) of Fig. 3 the frequency response considering the effects of the externally applied potential while keeping the ion concentration of the bulk fixed at $c_b = 10^{-3}$ M for two values of the Stern thicknesses. While the curves showing the magnitude and phase responses in Figs. 3c and 3d preserve the capacitive behavior and resistive behavior at low-frequency and high-frequency regimes, they tend to cluster close together as the applied voltage ranges from the neutrality $V = 0$ to $V = 0.5$ V. Similar to Figs. 3a, 3b, by increasing the bias voltage from $V = 0$ to $V = 0.5$ V the impedance behavior shifts to lower frequencies where the low-frequency capacitive behavior sets in ($\hat{\phi} \rightarrow -90^\circ$ in the phase plot). The curves show insignificant effects of the Stern layer at low voltages, $V \lesssim 0.2$ V, but at higher voltages, the presence of Stern layer causes a shift of the curves to lower frequencies, thereby delaying the onset of the capacitive behavior.

Charging time.—For many related applications, such as micro-capacitors and battery electrodes,^{56,57} the least required time to a complete discharge with an efficiency of more than 50% is regarded as a primary criterion. This is linked to the impedance behavior of the cell when changed from entirely resistive (R) to entirely capacitive (C) and it occurs when the real part of the impedance is equal to the imaginary part of the impedance corresponding to the phase angle of $\hat{\phi} = -45^\circ$. The associated frequency is known as the characteristic frequency (RC frequency) and is obtained in our work in the scaled form as $\hat{\omega}_{RC} = \frac{1}{2}\sqrt{c_b}/C(\bar{v}, c_b)$. Figure 4 shows the variations of the characteristic frequency $\hat{\omega}_{RC}$ in the range of ion concentrations from 10^{-5} M to 10^{-1} M for two limiting cases of Stern thickness, $\hat{\alpha} = 10$ and $\hat{\alpha} = 0$, when subjected to different values of the bias voltage. According to the figure, low-bias voltages result in lower characteristic frequency, which does increase as the ion concentration c_b increases. In this range of the applied potential values, the effects of the Stern thickness are relatively weak. However, for high-bias voltages, we observe a higher characteristic frequency, thereby a lower charging time, with the effects of ion concentration being gradually diminished, such that the characteristic frequency remains almost constant for all values of c_b . The Stern layer exhibits stronger effects in the range of high-bias voltages, so that charging times are lower (higher $\hat{\omega}_{RC}$) when the Stern layer is absent, $\hat{\alpha} = 0$.

Sensitivity.—The sensitivity of the proposed graphene-based electrochemical system is another important aspect of this study, which refers to the ability of an electrode to detect small changes in the analytes (ions) based on changes in the components of the impedance such as phase and magnitude.⁵⁸ In this study, we use Bode phase response, $\hat{\phi}$, to evaluate the sensitivity of the graphene electrode, as opposed to Bode magnitude response, $|\hat{Z}|$, because it can provide a more accurate and stable measure of the sensor response to the analytes (ions) in experimental measurements. Phase angle impedance sensors are commonly used in various applications, including bioimpedance analysis, food quality assessment, and chemical concentration measurements.^{58,59}

The angular phase sensitivity denoted by $S_{\hat{\phi}}$ is defined according to Ref. 58 by

$$S_{\hat{\phi}} = \frac{\hat{\phi} - \hat{\phi}_{\text{ref}}}{\hat{\phi}_{\text{ref}}} \quad [79]$$

which is the relative change of phase with respect to a reference phase, $\hat{\phi}_{\text{ref}}$, evaluated at a reference ion concentration, c_{ref} . In the following, we study the dependence of the phase sensitivity on ion concentration for different values of the applied DC voltage, V , reference ion concentration, c_{ref} , and the operating frequency, $\hat{\omega}$, as illustrated in Fig. 5.

Figure 5a shows the phase sensitivity of the system subjected to different bias voltages at the operating frequency of $\hat{\omega} = 10^{-3}$ when the reference ion concentration is set $c_{\text{ref}} = 10^{-3}$ M. According to the figure, increasing the bias voltage boosts the sensitivity, which is lower in the presence of the Stern layer. The effects of the Stern layer are noticeable only as the applied voltage increases from $V = 0.2$ V to $V = 0.5$ V, while they are negligible at the neutrality $V = 0$ and $V = 0.1$ V. Furthermore, we note that the sensitivity of the system is significantly influenced by our choice of the rion

concentration, $c_{\text{ref}} = 10^{-3}$ M. Namely, at low ion concentrations, $c_b < 10^{-3}$ M, the sensitivity increases as c_b decreases, while the system is insensitive to variations of the ion concentration when it is higher than $c_b = 10^{-3}$ M.

Figure 5b shows the phase sensitivity subjected to different reference ion concentrations when the applied voltage is set $V = 0.2$ V at the frequency of $\hat{\omega} = 10^{-3}$. In the range of the reference concentrations from $c_{\text{ref}} = 10^{-1}$ M to $c_{\text{ref}} = 10^{-4}$ M, we see a positive sensitivity at low ion concentrations, $c_b \lesssim 10^{-3}$, whereas sensitivity is negligible for $c_b > 10^{-3}$, similar to Fig. 5a. The sensitivity magnitude is decreased when c_{ref} decreases in that range of reference concentrations, showing no discernible effects due to the presence of the Stern layer. A distinct behavior is observed at the lowest reference ion concentration, $c_{\text{ref}} = 10^{-5}$ M, where a negative sensitivity is acquired in the range of ion concentrations 10^{-5} M $< c_b \lesssim 10^{-2}$ M, while it slowly approaches zero as the concentration increases to $c_b = 10^{-1}$ M. Furthermore, we see a significant reduction in the magnitude of the sensitivity by the presence of the Stern layer at this reference concentration for ion concentrations 10^{-5} M $\lesssim c_b \lesssim 10^{-3}$ M.

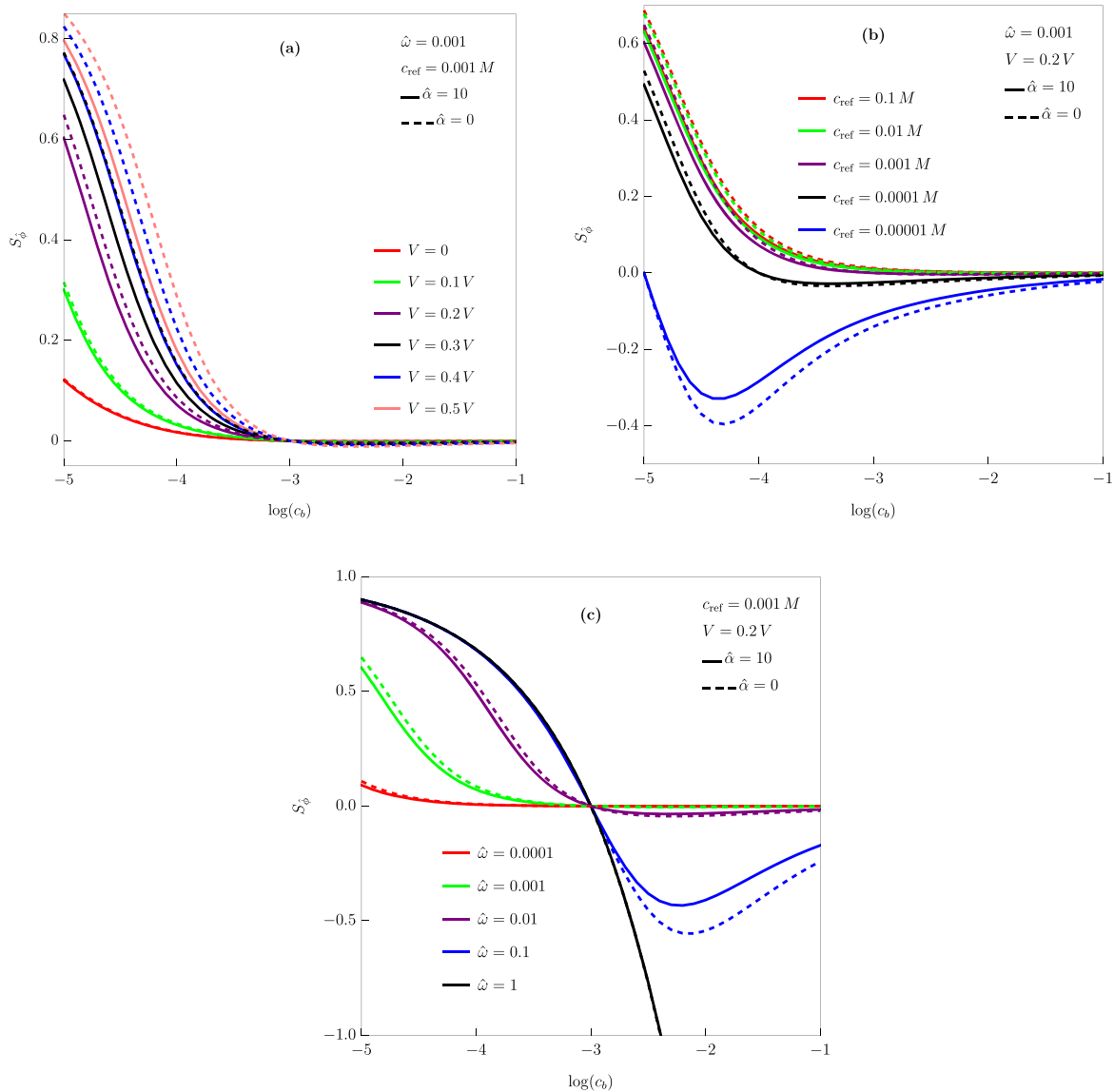


Figure 5. The phase sensitivity of the proposed graphene-based electrochemical cell vs ion concentration. (a) The variations of the phase sensitivity for different values of the bias voltage and the Stern thickness when the reference ion concentration and the operating frequency are set to $c_{\text{ref}} = 10^{-3}$ M and $\hat{\omega} = 10^{-3}$. (b) The variations of the phase sensitivity for different values of the reference ion concentration and the Stern thickness, when $V = 0.2$ V and $\hat{\omega} = 10^{-3}$. (c) The variations of the phase sensitivity for different values of the operating frequency and the Stern thickness when $c_{\text{ref}} = 10^{-3}$ M and $V = 0.2$ V.

In Fig. 5c, we examine the behavior of the sensitivity for different operating frequencies when the reference ion concentration and bias voltage are fixed at $c_{\text{ref}} = 10^{-3}$ M and $V = 0.2$ V, respectively, as the Stern thickness varies. According to the figure, we observe two distinct behaviors in sensitivity on both sides of the ion concentration $c_b = 10^{-3}$ M. For weakly concentrated electrolytes, 10^{-5} M $< c_b < 10^{-3}$ M, we notice a positive sensitivity that increases as the working frequency increases from $\hat{\omega} = 10^{-4}$ to $\hat{\omega} = 10^{-1}$, but it shows no further variation between $\hat{\omega} = 10^{-1}$ and $\hat{\omega} = 1$. As the electrolyte becomes highly concentrated, 10^{-3} M $< c_b < 10^{-1}$ M, the system shows almost no sensitivity at low operating frequencies, between $\hat{\omega} = 10^{-4}$ and $\hat{\omega} = 10^{-2}$. However, an increased negative sensitivity is observed when the operating frequency is $\hat{\omega} = 10^{-1}$ with a drastic jump to a large negative sensitivity at $\hat{\omega} = 1$. Our calculations confirmed that essentially the same sensitivity curve as for $\hat{\omega} = 1$ is also obtained for higher operating frequencies, $\hat{\omega} = 10$, $\hat{\omega} = 10^2$, and $\hat{\omega} = 10^3$. Such behavior of the sensitivity may be attributed to the resistive behavior of the cell, which dominates at high frequencies. As for the effects of the Stern layer, we only observe a noticeable effect at the operating frequency of $\hat{\omega} = 10^{-1}$ for ion concentrations 10^{-3} M $< c_b < 10^{-1}$ M.

Conclusions and Future Directions

In this work, we have revealed the AC capabilities of a charged graphene-electrolyte interface of electrochemical systems in the limit of thin DL, $\varepsilon = \lambda_D/H \rightarrow 0$, using a matched asymptotic expansion of the PNP equations with the appropriate boundary conditions for a 1:1 symmetrical electrochemical cell, while taking into account the combination of the Stern and diffuse layers. The leading order asymptotic analysis is performed in the context of matched asymptotic expansion both in a fully nonlinear treatment of the equilibrium state for an arbitrary DC bias voltage V and in a linearized form of the time-dependent perturbation, where we have shown how the proper charging time scale, $\tau_c = \lambda_D H/D$, leads to a consistent result for the impedance of the cell.

Our analysis of the total capacitance in equilibrium showed that the presence of the Stern layer only becomes relevant at high bias voltages, $|V| \gtrsim 0.1$ V. By defining a scaled frequency $\hat{\omega}$ in a manner that is independent of the ion concentration in the bulk, c_b , our frequency-response analysis revealed that the low-frequency capacitive behavior, which is highly desirable for (bio)sensing applications, is pronounced by decreasing c_b and/or by raising the bias voltage. On the other hand, lowering $|V|$ and/or raising c_b promotes the high-frequency resistive behavior of the cell.

Observations based on a characteristic frequency, which separates the capacitive and the resistive behaviours of the cell, reveal that lowering both the bias voltage and the ion concentration results in longer charging times, while increasing the bias voltage leads to shorter charging times with little effects on the ion concentration. The presence of a Stern layer reduces the characteristic frequency at high-bias voltages, giving rise to longer charging times, but its effects are negligible at low-bias voltages.

Our investigation into the phase sensitivity of the graphene-based electrodes has led us to conclude that a rise in the bias voltage boosts sensitivity at low ion concentrations c_b , but the presence of the Stern layer hinders it. Furthermore, we have shown that the suggested graphene-based electrode is sensitive at low operating frequencies between $\hat{\omega} = 10^{-3}$ and $\hat{\omega} = 10^{-1}$ for c_b below the reference concentration c_{ref} . At the operating frequency of $\hat{\omega} = 10^{-1}$, we also observe an increased negative sensitivity for $c_b > c_{\text{ref}}$ with a noticeable hindrance by the Stern layer. We also show a drastic jump to a large negative sensitivity in that range of ion concentrations at frequencies exceeding $\hat{\omega} = 1$, which may be related to the onset of the resistive behavior of the cell.

Moreover, the properties of the obtained impedance based on the PNP model, particularly the low-frequency capacitive behavior,

high-frequency resistive behavior, and the characteristic frequency related to charging time can also be generalized to other well-known systems such as asymmetric electrochemical cells,⁶⁰ ionic-electronic mixed conductors,⁶¹ and ion-exchange membrane systems.⁶²

In the future, we shall study EIS of ionic liquid (ILs) in the presence of a graphene electrode close to the PZC based on the 4th-order modified Poisson-Boltzmann equation that arises from the Bazant-Storey-Kornyshev (BSK) model.⁶³ This can be achieved by linearizing BSK model for small surface potentials, examining different additional boundary conditions⁶⁴ and highlighting the role of quantum capacitance in the total capacitance of the graphene-IL interface in equilibrium. The outcome can aid engineers in designing a new generation of fast supercapacitors with nanoelectrodes that are essential for stable functioning of all renewable energy sources.

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